Technical Document: A Methodology for Distributing Personal Income

Marina Gindelsky
DistributionofPersonalIncome@bea.gov

The distribution of personal income is a topic of widespread interest, including for policymakers, economists, and statisticians. Though literature on the topic has dramatically increased over the past two decades, led particularly by Piketty and Saez (2003), income inequality has long been a topic of interest (Kuznets 1941, 1953). Indeed, the predecessor of the U.S. Bureau of Economic Analysis (BEA), the Office of Business Economics, published estimates of the "Size Distribution of Income" covering select years between the mid-1940s and the 1970s (Office of Business Economics 1953).

Recently, interest has grown in the distributional national accounts. As each household member has some claim on the economic resources of the household and affects decisions regarding economic activities, the household is regarded by the U.S. Census Bureau and in the System of National Accounts (2008) as the institutional unit for compiling distributional results (Organisation for Economic Co-operation and Development 2018). When households are used as the unit of measurement and national accounts as the aggregate, personal income represents a natural and appropriate income concept for decision-making (Fixler, Gindelsky, and Johnson 2019). Personal income is the income received by, or on behalf of, all persons from all sources—from participation as laborers in production, from owning a home or business, from the ownership of financial assets, and from government and business in the form of transfers. It includes income from domestic sources as well as from the rest of the world; it does not include realized or unrealized capital gains or losses.

There are many metrics that can be used to measure the distribution of personal income and quantify income inequality. A full set of metrics for 2023 is available on the BEA

website under "National Data & Documentation" in an Excel file (see sheets "Table 1" and "Table 2"), with a partial set of provisional metrics for 2024 in a separate file. Table 1, "Major Components of Personal Income and Disposable Personal Income by Decile," distributes the household income portion of National Income and Product Accounts (NIPAs) table 2.9, "Personal Income and Its Disposition by Households and by Nonprofit Institutions Serving Households," by decile in two ways. First, households are ranked by equivalized (household size-adjusted) personal income and assigned to deciles (the first 10 numerical columns). Next, households are re-ranked by equivalized disposable personal income (personal income less taxes) and assigned to deciles. Table 1 also includes totals for each line item in the first column, along with summary lines for personal income and disposable personal income.[1] The deciles of personal income and disposable personal income are published as iTable 2.10.

By providing shares of income by both personal income decile and disposable personal income decile, we are able to glean additional distributional insights, both as compared to the commonly provided quintiles (Census tables A3 and A4) and relative to solely pretax income. Table 2, "Inequality Metrics," reports inequality statistics for personal income and disposable personal income.[1]

A summary file containing all available metrics for all available years has been uploaded for ease of comparison of concepts and quantiles across the time series. This broad set of statistics provides a comprehensive overview of the distribution of income, including the share in each quintile, the top 1 percent share, the top 5 percent share, the Gini index, and income ratios.[2] These statistics are featured in the literature (including Piketty, Saez, and Zucman 2018; Atkinson, Piketty, and Saez 2011; Auten and Splinter 2019) as well as the World Inequality Database, Federal Reserve Board accounts (Distributional Financial Accounts), the Census Bureau (aforementioned tables A3 and A4) and U.S. Bureau of Labor Statistics (BLS) publications (Cunningham 2015).

Below is a description of the methodology used to estimate the distribution of personal income, as found in the NIPA tables, specifically table 2.1, "Personal Income and Its Disposition" line 1. This methodology also underlies the construction of the internationally comparable OECD research estimates (see accompanying OECD technical

document). Section 1 describes the overall strategy, including how external data sets are used to calculate the statistics. Sections 2 through 7 outline the major components of personal income. Section 8 explains the extrapolation which underlies the provisional results and Section 9 describes technical updates to the previous release. Section 10 summarizes the data sources.

1. Overall Strategy

The Annual Social and Economic Supplement of the Current Population Survey (a base microdata set, hereafter referred to as "CPS") was used for the analysis. Many national accounts components must be allocated to households (that is, each household is assigned a proportional share of each component). These components can be categorized in four broad categories under "Decomposition of Personal Income for Households," on the BEA website. These include: (1) adjusted money income (AMI), (2) financial items (F), (3) health items (H), and (4) other transfers (net) (T). Some of these components were allocated to households using additional data in conjunction with CPS variables. To maximize transparency and utility to data users, most data used were public-use data and may be accessed using the links provided in section 10, "Data Sources."

The CPS, jointly sponsored by the Census Bureau and BLS, is a nationally representative annual household survey of the civilian noninstitutionalized population (approximately 92,000 households sampled, representing 130 million households for 2020) that includes many demographic and income questions and is administered in February, March, and April of each year. According to the Census Bureau, the survey is the primary source of labor force statistics for the United States.[3] It is important to note the income data in the CPS are 1 year behind the data collection year (that is, the survey year). Our analysis starts in 2000 due to the annual availability and comprehensiveness of some component data sources (see section 10).

Overall, the estimation strategy consists of the following four steps: (1) identify a NIPA total to be distributed, (2) identify CPS variable(s) and external variables that could be used to allocate this total, (3) sum all component NIPA totals to personal income and subtotals of interest, and (4) construct inequality statistics. This process yields a data set

in which each CPS household has a value for each component of personal income. After all components have been summed to compute personal income, equivalized personal income is calculated by dividing personal income by the square root of the number of household members. For example, if household income is $10,000 and there are four members of the household, equivalized household income is $5,000 (half of $10,000). Equivalization allows incomes to be compared across households of different sizes. Equivalized rankings of income are used for all income inequality metrics.

2. Allocations based on external data sets

Several additional data sets are used to allocate NIPA totals to households.

A. Adjustment for very high incomes

Data from the Internal Revenue Service (IRS) Statistics of Income (SOI) program are used to adjust the top of the income distribution (that is, highest incomes) for three main reasons. First, the CPS is believed to unsuccessfully survey those with very high incomes, introducing nonresponse bias in inequality estimates (Bollinger et al. 2018). Second, there is a perception that there is underreporting by those with top incomes. Third, the CPS has top codes, which vary by year, for those with top incomes so as not to risk identification of those individuals. For example, if an individual reports an income of $10 million annually, they may instead be assigned a value of $1 million by Census. For these reasons, it becomes prudent to adjust CPS incomes (Armour et al. 2021). The adjustment process is described below.[4]

First, we account for misreporting in the published SOI data. As discussed in detail by DeBacker et al. (2020), Johns and Slemrod (2010), Auten and Splinter (2019), and others, audit data from the IRS National Research Program and Taxpayer Compliance Measurement Program have shown that administrative data are significantly underreported (i.e., the existence of a "tax gap") due to tax noncompliance. BEA does make an adjustment for aggregate misreporting of proprietors' income and partnership income in the macrodata, seen in NIPA table 7.14, line 2. However, the extent of misreporting differs substantially by income source (for example, self-employment vs.

wage income) and varies significantly across the distribution.[5] Thus, we first multiply the SOI data for wages, self-employment, dividends, and interest by their respective adjustment factors to be consistent with the audited totals.[6]

Next, we aggregate SOI data on tax units (for 2024, internal tax data are used—see section 8) into the relative proportions of income with adjusted gross income (AGI) in categories as follows: less than $200,000, $200,000–$500,000, and AGI greater than or equal to $500,000.[7] These categorizations are done for four components:[8] (1) wages, (2) self-employment income (including farm), (3) dividends, (4) interest income, and (5) rents and royalties. For each tax year, these shares sum to 1 across income categories. For example, a 2022 CPS that asks respondents about 2021 income would correspond to tax year 2021 SOI data. For each of the five components listed above (simply referred to as Source X below), the procedure is as follows:

1. Create tax units[9] in the CPS and sort by AGI into three groups as above:
    - AGI greater than or equal to $500,000[10]
    - AGI of $200,000–$500,000
    - AGI less than $200,000
2. Allocate these totals to CPS tax units (which we construct) corresponding to the above categories such that the resulting distribution resembles the SOI data, respectively for each Source X.

For example, if the SOI distribution of dividends was calculated such that tax units with AGI under $200,000, $200,000–$500,000, and $500,000+ had shares of total dividends of 50%, 30%, and 20% respectively, tax units in the CPS are allocated shares of the NIPA dividend total such that those with AGI under $200,000, $200,000–$500,000, and $500,000+ end up with (weighted) shares of total NIPA dividends of 50%, 30%, and 20% respectively.

B. Adjustment for underreporting of certain government programs

In the current exercise, we model the recipiency of Supplemental Security Income (SSI), Medicaid, and Supplemental Nutrition Assistance Program (SNAP) benefits to each individual in the CPS for a given survey year. From administrative data on enrollment in these programs available from the Social Security Administration, Kaiser Family Foundation, and U.S. Department of Agriculture, respectively, we first identify the gap in underreporting. We then use the general approach of the Congressional Budget Office (CBO) (Habib 2018) to determine who in the CPS is most likely to underreport program participation and use a machine learning method to improve the identification of those individuals. More specifically, we use a Random Forest algorithm, which can model varied nonlinearities in the data. Program participation probabilities are estimated using out-of-sample predictions to avoid overfitting bias. Finally, we impute program income for these individuals, such that it corresponds to state-level NIPA totals.[11]

We also adjust for underreporting of unemployment income for 2000–2021 using a crosswalk from Larrimore, Mortenson, and Splinter (2023). This adjustment imputes recipiency to CPS respondents and assigns values, which we then scale to the NIPA total.

C. Imputation of financial income

We use the Summary Extract Public Dataset of the Survey of Consumer Finances (SCF), as obtained from the Federal Reserve Board of Governors Economic Research website.[12] We use this data set to distribute the three imputed components of personal income, which are described for each relevant component. Because the SCF is triennial, we first interpolate the SCF variables for the years in which the SCF is not observed.[13] We match the SCF survey year to the CPS survey year (both income variables are lagged 1 year, while asset variables are contemporaneous). First, households are binned by income (not including capital gains). These eight bins are constructed analogously in the CPS. In order to allocate imputed interest received from banks, credit agencies, and investment companies, we first calculate the total amount of "banking" (checking plus savings balances) and "insurance" (cash value of life insurance). Then, the relative share of each

income bin to total income is imputed for each bin and matched to households in the CPS by corresponding income bin. Thus, the NIPA totals for imputed interest for these two categories are then allocated proportionally by income.

To allocate imputed interest[14] and dividends from pensions, we first calculate the shares of retirees with defined contribution (DC) and defined benefit (DB) plans, respectively. Next, we calculate the share of DC participants who also have DB plans and the share of disbursements that comes from DC plans. Prior to the CPS survey redesign, the relevant information is (1) whether an individual is participating in a retirement plan or (2) whether an individual receives disbursements from a retirement plan. It is not reported in the CPS whether an individual is participating in a DC or DB plan explicitly. Thus, individuals reporting participation in a retirement plan are randomly allocated to DC alone, DB alone, or DB and DC, until the distribution of plan participation matches that observed in the SCF. Once individuals have been allocated to DC, DB, or both, they are given a share of the imputed interest and dividends. Similarly, a portion of those reporting "Regular payments from IRA, KEOGH, or 401(k) accounts" is assigned to the group of those receiving disbursements from DC plans, consistent with the share in the SCF. It is possible to reasonably identify those receiving disbursements from DB plans in the CPS through that same question. Recipients are given a share of the imputed interest and dividends according to their reported disbursement amount.

After the CPS survey redesign (implemented in 2018), it is possible to identify those contributing to retirement plans and those receiving disbursements from DC plans. It is, however, not possible to identify those participating in both DB and DC plans. Thus, participants in DB plans are allocated a share of imputed interest and dividends according to their wage, while participants in DC plans are allocated a share according to their reported contribution. Those receiving disbursements are allocated a proportional share.

D. Imputation of rental income for owner-occupied housing

We use the Consumer Expenditure Survey (CE) from BLS to impute rental income for owner-occupied housing for 2000–2004. Using this data source, we first rank "consumer units" (roughly equivalent to households) by before-tax income, creating deciles. We

next construct a share of rental equivalence to before-tax family income. For example, if a household's income is $100,000 and it reports the expected rental value of $4,000 per month ($48,000 annually), the rent-to-income share would be 48 percent.

The median share is calculated for each income decile. This share is then applied to income deciles (ranked by money income) in the CPS for owner-occupied households to impute a value of rental income based on their income.

We use the American Community Survey (ACS) from the Census Bureau in a similar way as the CE to impute rental income for owner-occupied housing for 2005 forward. The ACS is preferred given the sample size of the survey. Though the ACS does not contain a rental equivalence question, we use the BEA strategy of Rassier et al. (2021) to impute a value for each household. We use stacked 5-year Public Use Microdata Sample (PUMS) files where possible to have the largest possible sample size and 1-year files where not. ACS households are ranked on money income (as closely as possible to the CPS definition), and then sorted into before-tax deciles. Then, the share of rental equivalence is calculated as above for the CE.

E. Employer contribution to health insurance and Medicare

In the latest CPS redesign, the Census Bureau no longer provides an imputation for the employer contribution to health insurance. Accordingly, we directly impute this variable for 2010 onward. We use the MEPSnet Query Tool to create tables of average premium per enrolled employee and percent of total premiums contributed by employees by family size (single or family), state, year, and firm size at private sector establishments. If CPS respondents report that the employer pays the full premium, the average employer premium is imputed by state, year, family size, and firm size for the respondent. If CPS respondents report that the employer pays part of the premium and the amount paid out of pocket, the employer contribution is then imputed using the average percent of the premium that employees pay by state, year, family size, and firm size.

Additionally, we use the Medicare 5% microdata to calculate the average individual

Medicare expenditure by year, age category, and disability status. This is then imputed to each CPS individual reporting Medicare recipiency and then scaled to match the NIPA totals (by state). Although these nonpublic data are provided to BEA by the Centers for Medicare & Medicaid Services (CMS), we have compiled a state-level crosswalk, available upon request. We have switched to this data source to extend the series back to 2000 and create a more nuanced imputation. The previously used publicly available CMS average expenditure data files were unavailable before 2007.

As noted above, the following sections will describe the core aggregated income components. A numeric breakout by relative size is available for the December 2020 release in Fixler, Gindelsky, and Johnson 2020.

3. Adjusted Money Income (AMI)

In these estimates, we adjust money income (as defined by Census) to be consistent with the concepts used in the NIPA estimates. It is helpful to compare the definitions of Census Bureau money income and personal income:

"Census money income is defined as income received on a regular basis (exclusive of certain money receipts such as capital gains) before payments for personal income taxes, social security, union dues, Medicare deductions, etc. Therefore, money income does not reflect the fact that some families receive part of their income in the form of noncash benefits, such as food stamps, health benefits, subsidized housing, and goods produced and consumed on the farm." (From "About Income & Poverty" on the Census Bureau website)

"Personal income is the income that persons receive in return for their provision of labor, land, and capital used in current production, plus current transfer receipts less contributions for government social insurance (domestic). Personal income is equal to national income minus corporate profits with IVA and CCAdj, taxes on production and imports less subsidies, contributions for government social insurance, net interest and miscellaneous payments on assets, business current transfer payments (net), and current surplus of government enterprises, plus personal income receipts on assets and personal current transfer receipts." (From Chapter 2 of Concepts and Methods of the U.S. National Income and Product Accounts.)

Though Census Bureau money income, in many ways, is a narrower definition of income, it does include variables that are not in personal income, such as retirement disbursements. Accordingly, we add up the components of Census Bureau money

income that are included in personal income, excluding variables such as retirement disbursements (*dbtn-val*) or certain sources of disability income (*dis-val*) and survivor income (*sur-val*).[15] We call this approximation adjusted money income.[16] It is primarily comprised of income from wages and salaries, self-employment (farm and nonfarm), interest, dividends, and social security income, which together sum to 95 percent of adjusted money income in 2018 (a representative year). The remaining 5 percent is comprised of income from additional sources such as rents and royalties, unemployment insurance, and disability income, among others.

We distribute the components that make up AMI individually. Wages and salaries, farm and nonfarm income, interest income, and dividend income are distributed by the relevant variables, with the SOI adjustment referred to in section 2A. Federal benefits, including social security, unemployment insurance, railroad retirement, black lung benefits, pension benefit guaranty, veterans' benefits, and workers' compensation are distributed by the relevant variables in the CPS.

Unemployment insurance has been imputed for 2000–2021 using a crosswalk from Larrimore, Mortenson, and Splinter (2023), which corrects for the underreporting of such payments in the CPS; the crosswalk is based on administrative data. This is particularly influential for 2020–2021 estimates.[17]

State and local benefits including temporary disability insurance, public assistance, employment and training, education, and others are distributed to CPS households by their relevant variables, respectively. Household current transfer receipts from nonprofit institutions are distributed to CPS households by the value of their educational assistance in the CPS.

4. Financial Items

This section describes the imputations for financial items in the NIPAs to CPS households. This category includes allocations for pensions and profit sharing, life insurance, rental income from owner-occupied housing, and imputed interest and dividends.

Employer contributions to pension plans and group life insurance are distributed to those who participate in employer-sponsored pension plans and health insurance, respectively, by wage. Imputed interest and dividends are distributed using an imputation derived from the SCF for banking, insurance, and retirement assets for each relevant item, as referred to in section 2C. Owner-occupied housing rental income is distributed using an imputation derived from the CE (2000–2004) and the ACS (2005 forward) by income bracket referred to in section 2D.

5. Health Items

This section describes the methods used to impute health items in the NIPAs to CPS households. This category is the sum of allocations for employer contributions for health insurance, Medicare,[18] Medicaid, military medical insurance, and other medical care payment assistance.

Employer contributions to health insurance are distributed by the corresponding CPS variable until 2010. After 2010, employer contributions to health insurance are distributed using data from MEPS, as described in section 2E. The exception is military medical insurance, which is distributed to active military members.

As our objective is to distribute the components of personal income, we must distribute the expenditure on Medicare and Medicaid to participants. Medicare is distributed by assigning average individual expenditures by year, age category, and disability status using Medicare 5% data, as described in section 2E. Medicaid is distributed using a machine learning approach that assigns program receipt and imputes values consistent with NIPA state-level totals, as described in section 2B. Medical assistance is distributed to those who are assigned SNAP and report Women, Infants, and Children (WIC) benefits or other assistance, except the Children's Health Insurance Program (CHIP), which provides assistance to households reporting children covered by CHIP.

6. Net Other Transfers

This section describes the methods used to impute items related to transfers in the NIPAs that are not contained in adjusted money income or health to CPS households. These items include (net) employer and employee contributions for government social insurance,[19] SNAP, WIC, refundable tax credits, energy assistance, educational assistance, and other transfers.

Employer/employee contributions to old-age, survivors, disability, and hospital insurance are distributed by the imputed value of FICA (Federal Insurance Contributions Act) as calculated by TAXSIM-35 (and subsequently PolicyEngine) Employer contributions to workers' compensation and supplemental unemployment are distributed respectively by wage. Other employer and employee/self-employed contributions to government social insurance are also distributed by wage. Military medical insurance (federal benefits and employee/self-employed contributions to government social insurance) are distributed to CPS households with active military members. SNAP is distributed using the approach described in section 2B.

Refundable tax credits are distributed to CPS households separately by the criteria for the relevant credit where possible. For example, the CPS values for the earned income tax credit (EITC), child tax credit (CTC), and additional child tax credit benefits are used, as is participation in the marketplace for the health premium tax credit.[20] Together, these categories represent 95 percent of the refundable tax credits in 2018. One complication is that the tax simulator calculations impute the credits for the previous calendar year, regardless of when they are received by the households. Thus, there can be an issue of timing for some credits and households.

Energy assistance and WIC are distributed to CPS households by the corresponding CPS variables. All other transfers are distributed to households reporting receipt of WIC, SNAP, or other assistance. Finally, "household current transfer receipts from business (net)" are distributed according to the distribution of all other (summed) components of personal income. As there is no other information available to distribute this income component, it is distributed so that its inclusion does not meaningfully affect the overall

income distribution.[21]

There are some items in the NIPA accounts for which no distributional information is available. These items are distributed such that their inclusion does not change the overall distribution. For example, if income consisted of wages, self-employment, and other, and the Gini of the "wages and self-employment" total were 0.5, the "other" category is now distributed according to the total of "wages and self-employment" and thus the Gini remains at 0.5. One such imputed total is "Household current transfer receipts from business" in Table 1 (line 9) for each year.

To calculate personal income from household income, household current transfer receipts from nonprofits and nonprofit institution transfer receipts from households are deducted, and nonprofit institution income is added. As with "household current transfer receipts from business," this residual is distributed such that it does not meaningfully change the income distribution.

7. Taxes

The calculation of tax liability (and associated components) was done using the NBER TAXSIM-35 program (Feenberg and Coutts 1993) for 2000–2020. CPS households were organized into tax units, whose income inputs are scaled to NIPA totals. Federal taxes before credits (*fiitax*), state taxes before credits (*siitax*), FICA, EITC, CTC, and alternative minimum tax liability were merged from TAXSIM-35 back into the CPS for the calculations to proceed.

Support for TAXSIM-35 has been discontinued as of 2025. PolicyEngine, a nonprofit, has created a high-fidelity emulator to enable data users to continue computing TAXSIM-35 outputs. The creation of this emulator has enabled BEA to continue to produce Disposable Personal Income. It is available here.

In the latest CPS redesign, the Census Bureau imputation for property taxes is no longer available. Thus, we impute property taxes by calculating the property tax rate using the ratio of tax to property value in the ACS by public use microdata area by county. When a local rate is unavailable due to disclosure restrictions, we impute a state-level property

tax rate. Disposable personal income is calculated by subtracting federal and state taxes from personal income.

8. Provisional Estimates

As in prior releases, BEA provides official estimates with a 2-year lag and provisional estimates with a 1-year lag. We utilized 1040 microdata provided to the Census Bureau to evaluate and forecast key income sources. Specifically, we compared the distributions for AGI, wages, interest, and dividends for the full tax year (as received by Census) across multiple years.

The uncertainty accompanying the provisional estimates for 2024 is largely a function of this extrapolation, as tax data are the key piece of information used for adjusting the CPS for underreporting (as described in section 2A). Notably, tax information on partnerships and sole proprietorships from SOI was unavailable at the time of estimate construction. This information is used to derive estimates for nonfarm proprietors' income, a key component of personal and national income for 2023. This may potentially lead to a larger future revision for 2023.

Thus, significant deviations in the shares of top incomes in the tax data lead to movements in the top shares of the distributional estimates. Accordingly, we have provided a range for the top 1 percent and top 5 percent shares, rather than exact point estimates. The ranges provided reflect analysis conducted with historical data.

This extrapolation strategy was first tested in the December 2022 release. We found that incorporation of the published SOI data did lead to top share estimates falling within the published ranges, unless there was a methodological change. Therefore, this strategy was also utilized for the present release, with the understanding that though changes in top shares may lead to changes in decile shares as well, analysis has shown most of the expected variation to be within the top decile. At present, we do not expect large revisions to these shares (independent of significant updates to NIPA totals or methodology) which fall outside of these intervals.

9. Technical Updates

There have been a few updates since the previous release (December 2024). We have further refined the technical methodology for adjusting top incomes. First, we implemented a coding update which improved the accuracy of assigning households to the three AGI categories discussed in section 2A. Second, we have improved the allocation of dividends by utilizing internal BEA tables to refine the disaggregation by source. The net impact of these improvements was to slightly reduce inequality for most years (on average -0.4pp for the top quintile). The most recent years were more heavily impacted by NIPA revisions.

10. Data Sources

There were many data sources (listed below with links) used for the compilation of these estimates. Most are publicly available and made available with varying time lags after the data are collected. Information on incomes from the CPS and CE is regularly available with a 1-year time lag since the surveys ask respondents about income received in the previous calendar year. The SCF is a triennial survey and asks about income received in the previous year, similar to the CPS and the CE. Medical Expenditure Panel Survey (MEPS) published estimates are available with a 1-year lag, while the microdata are available with a 2-year lag.

1. **National Income and Product Accounts.** The primary source for our national macrodata for 2000 to 2024 is the BEA NIPA data. These data are subject to frequent revision. The current set of estimates uses the January 2026 release.

2. **Current Population Survey, March Supplement.** This is the core data series that is the base of the analysis for all noninstitutionalized households for survey years 2001 to 2025. These data were downloaded from the NBER website for 2000 to 2017. For 2018–2025, they were used directly from the Census Bureau website.

3. **Consumer Expenditure Survey.** This survey was used to estimate rental equivalence for owner-occupied housing from 2000–2004. The data were downloaded from BLS.

4. **Survey of Consumer Finances.** This survey was used to distribute imputed interest income from the NIPAs for 2007, 2010, 2013, 2016, 2019, and 2022. For the non-survey years, values were imputed with the methodology described in section 2C. The data were downloaded from the Federal Reserve Board.

5. **Internal Revenue Service Statistics of Income data.** In order to adjust top incomes in the CPS, summary IRS Form 1040 data are aggregated by AGI for tax years 2000–2023 in table 1.4 found in "SOI Tax Stats" on the IRS website.

6. **American Community Survey 2005–2019 (stacked 5-year PUMS files) and 1-year files for 2020 forward.** In order to impute property tax rates at the county level, stacked 5-year files are used where available, and 1-year files for the most recent year.

7. **Medical Expenditure Panel Survey data.** Tables II.C.1, II.C.3, II.D.1, and II.D.3 were downloaded (2010–2024) using the MEPSnet Query for the insurance component to impute the employer premium and percent employee contribution.

8. **Centers for Medicare & Medicaid Services.** Medicare 5% microdata were provided to BEA by CMS and are used to perform calculations for Medicare expenditure, at the state-level (by age and disability status).

9. **IRS 1040 Microdata.** 1040 microdata made available to the Census Bureau by the IRS are utilized to calculate provisional distributional results for 2024. These data include all tax units who have filed returns for the 2024 tax year. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product (Data Management System (DMS) number: P-7503840, Disclosure Review Board (DRB) approval number: CBDRB-FY26-SEHSD003-041).

10. **Unemployment.** To address underreporting of unemployment in the CPS, a crosswalk (2000–2021) from Larrimore, Mortenson, and Splinter (2023) is employed. This crosswalk provides imputed recipiency and benefits to CPS individuals based on administrative data.

References

Armour, Philip, Gerald Auten, Richard Burkhauser, and Jeff Larrimore. 2021. "Recent Trends in U.S. Top Income Shares in Tax Record Data Using More Comprehensive Measures of Income Including Accrued Capital Gains." *Journal of Political Economy* 129 (5): 1319–1360.

Atkinson, Anthony B., Thomas Piketty, and Emmanuel Saez. 2011. "Top Incomes in the Long Run of History." *Journal of Economic Literature* 49 (1): 3–71.

Auten, Gerald, and David Splinter. 2019. "Top 1 Percent Income Shares: Comparing Estimates Using Tax Data." *AEA Papers and Proceedings* 109 (1): 307–311.

Bollinger, Christopher R., Barry T. Hirsch, Charles M. Hokayem, and James P. Ziliak. 2019. "Trouble in the Tails? What We Know about Earnings Nonresponse 30 Years after Lillard, Smith, and Welch." *Journal of Political Economy* 127 (5).

Cunningham, J. Chris. 2015. "Measuring wage inequality within and across U.S. metropolitan areas, 2003–13." *Monthly Labor Review* 138.

DeBacker, Jason, Bradley Heim, Anh Tran, and Alexander Yuskavage. 2020. "Tax Noncompliance and Measures of Income Inequality." *Tax Notes* 166 (7): 1103–1118.

Feenberg, Daniel, and Elizabeth Coutts. 1993. "An Introduction to the TAXSIM Model." *Journal of Policy Analysis and Management* 12 (1): 189–194.

Fixler, Dennis, David Johnson, Kevin Furlong, and Andrew Craig. 2017. "A Consistent Data Series to Evaluate Growth and Inequality in the National Accounts." *Review of Income and Wealth* 63 (S2): S437–S459.

Fixler, Dennis, Marina Gindelsky, and David Johnson. 2020. "Measuring Inequality in the National Accounts." BEA Working Papers, WP 2020–03.

Fixler, Dennis, Marina Gindelsky, and David Johnson. 2019. "Improving the Measure of the Distribution of Personal Income." *AEA Papers and Proceedings* 109: 302–306.

Fixler, Dennis, Marina Gindelsky, and David Johnson. 2018. "Towards a Distribution of Household Income: Linking Survey Data to Administrative Data." Paper presented at the International Association for Research in Income and Wealth 35th General Conference, Copenhagen, Denmark, August 20–25.

Gindelsky, Marina. 2020. "Technical Document: An Updated Methodology for Distributing Personal Income." BEA website.

Gindelsky, Marina. 2023. "Technical Document: An Updated Methodology for Distributing Personal Income." BEA website.

Gindelsky, Marina. 2024. "Technical Document: An Updated Methodology for

Distributing Personal Income." BEA website.

Habib, Bilal. 2018. "How CBO Adjusts for Survey Underreporting of Transfer Income in Its Distributional Analyses: WP 2018-07." CBO Working Papers, July.

Johns, Andrew and Joel Slemrod. 2010. "The Distribution of Income Tax Noncompliance." *National Tax Journal* 63 (3): 397–418.

Kornfeld, Robert. 2023. "Improved Measures of Regulated Investment Companies and Real Estate Investment Trusts for the U.S. National Economic Accounts." *Survey of Current Business*. May 2023.

Kuznets, Simon. 1941. National Income and Its Composition, 1919–1938, Volume I. New York: National Bureau of Economic Research.

Kuznets, Simon. 1953. Shares of Upper Income Groups in Income and Savings. New York: National Bureau of Economic Research.

Larrimore, Jeff, Jacob Mortenson, and David Splinter. 2023. "Unemployment Insurance in Survey and Administrative Data," *Journal of Policy Analysis and Management* 42(2) pp. 571-579.

Mataloni, Lisa, Dorian Turner, and Stephanie McCulla. 2023. "Improved Estimates of the National Economic Accounts." *Survey of Current Business*. November 2023.

Organisation for Economic Co-operation and Development. 2018. Handbook on compiling distributional results on household income, consumption and saving consistent with national accounts.

Piketty, Thomas, and Emmanuel Saez. 2003. "Income Inequality in the United States, 1913–1998." *Quarterly Journal of Economics* 118 (1).

Piketty, Thomas, Emmanuel Saez, and Gabriel Zucman. 2018. "Distributional National Accounts: Methods and Estimates for the United States." *Quarterly Journal of Economics* 13 (2).

Rassier, Dylan G., Bettina H. Aten, Eric B. Figueroa, Solomon Kublashvili, Brian J. Smith, and Jack York. 2021. "Improved Measures of Housing Services for the U.S. Economic Account." *Survey of Current Business* 101 (5).

U.S. Office of Business Economics. 1953. Income Distribution in the United States, by Size, 1944–1950. Washington, DC: U.S. Government Printing Office.

[1] Results ranked by household income are available upon request.

[2] An appendix table with inequality statistics calculated based on equivalized income is available upon request.

[3] During the period of analysis here, there has been a significant redesign. We incorporated the first portion of the redesign in 2014 and the second in 2018. This redesign significantly affects the results, raising inequality.

[4] Previous exercises had different adjustment strategies. See: Fixler et al. 2017; Fixler, Gindelsky, and Johnson

2018, 2019, 2020; Gindelsky 2020, 2023, 2024.

[5] DeBacker et al. (2020) provide estimates (2006–2014) of misreporting by income category for wages, dividends, Schedule C, Schedule D, Schedule E, and "All Other" in table A4, "Percentage Changes in Income as a Result of Audit by Income Source and AGI Group."

[6] Although we are able to roughly adjust the SOI data to reflect audited totals for these categories, we are not able to do a detailed adjustment of the SOI data to reflect the higher "true" totals that would include underreported income that examiners were unable to detect (Johns and Slemrod 2010). IRS researchers employ an econometric technique called "detection-controlled estimation" (DCE) to model the true amounts. BEA received a report from IRS reflecting this "true" misreporting for proprietors' income and partnership income, which is used to adjust business income in NIPA table 7.14. However, though Johns and Slemrod (2010) do provide some distributional estimates of the misreporting impact (reported versus "true") in their tables, they do so for percentiles and for tax year 2001. Thus, we are unable to use these for our current exercise.

[7] We had slightly modified our household allocation coding in order to assign households to the tail. This has allowed us to return to 3 categories, compared to 2 for 2000-2010 in the previous release.

[8] We run TAXSIM-35 (a National Bureau of Economic Research (NBER) program that calculates tax liability from survey data) for 2000-2020 to impute tax variables based on the scaled NIPA values for wages, self-employment, and unemployment. One of the model outputs, AGI, is very highly correlated with the CPS AGI variable for the years after 2010 due to improvements in Census Bureau modelling. We do not have information on capital gains in the CPS for 2008–2017, which is used in TAXSIM-35, thus our estimated tax liability would be slightly understated for those years. Since TAXSIM-35 was discontinued in 2025, we run PolicyEngine for years 2021 forward. See section 7 for more.

[9] As defined by the Urban Institute, "A tax unit is an individual, or a married couple, that files a tax return or would file a tax return if their income were high enough, along with all dependents of that individual or married couple. A tax unit is therefore different than a family or a household in certain situations. For example, a cohabiting couple constitutes one household but if the individuals are not legally married, they would file separate tax returns and thus be considered two tax units. A family could consist of a married couple and the wife's elderly mother who lives with them. That family would be considered two tax units since, if the elderly mother had a large enough income, she would be required to file a federal income tax return on her own. In general, the number of tax units tends to be larger than the number of families or households reported elsewhere."

[10] Households are sorted into tax units, using information on household structure and filing status. If the proportion of tax units with AGI of at least $500,000 is slightly too low to match the proportion in the SOI tables, a subset of high earnings households is iteratively added to the group until this proportion is roughly equivalent to that of the SOI for the given year. We do the same with AGI of $200,000–$500,000. We do not do this for households with lower incomes, given that it may well be that households at the lower end do not file taxes but still report earned incomes to the CPS. Our approach thus becomes a weighted average of the CPS and SOI data. Tax units are re-aggregated into households afterward such that when aggregated across all households for *Source X,* the total will add up to the NIPA total.

[11] In preliminary versions of our exercise (Gindelsky 2020), we used a crosswalk, provided directly by CBO, that uses an algorithm to assign probabilities of receipt of SSI, Medicaid, and SNAP to each individual in the CPS for a given survey year and then correspondingly imputes values. As the CBO estimates are not available on a regular schedule (and when published, with a 2-year lag), we modeled the benefits to match as closely as possible. Our resulting overall recipiency and program income means are 99 percent correlated with those of CBO for 2000–2018. On a household level, program income recipiency and means are correlated in the microdata at more than 70 percent for all programs. However, we have greater household-level variance in our imputations due to our process of controlling to state-level NIPA program totals, available in NIPA table SAINC35.

[12] Additionally, we use imputations of defined benefit pensions available here in an SCF crosswalk, provided by the Fed.

[13] We use the Fernandez procedure in the R package "tempdisagg." The Fernandez procedure extends the Denton and Chow-Lin approaches by obtaining its solution by minimizing a quadratic loss function in the differences between the series to be created and a linear combination of the high-frequency series. The results of this method very closely match the results of the interpolation used in the Federal Reserve Board Distributional Financial Accounts.

[14] This imputed interest is deducted from the interest variable in the CPS when scaling monetary interest to the NIPA total.

[15] Prior to the 2014–2018 redesign, the relevant variable was "ret-val." The exclusion of retirement disbursements constitutes approximately 75 percent of the money income excluded.

[16] Adjusted money income also excludes potential sources of intrasectoral transfers, which would net out in the sector and are not associated with current period production, such as other financial assistance (*fin-val*), other income (*oi-val*), alimony (*alm-val*), child support (*csp-val*), and other noninstitutional educational assistance (*ed-val*). We do include incomes from these sources tied to railroad retirement, other retirement, workers' compensation, black lung benefits, and state and local government disability.

[17] Unemployment insurance from 2022 on is imputed according to the cps variable.

[18] The totals for social security, Medicare, and Medicaid have been adjusted slightly down for the proportion of likely nursing home residents, not surveyed by CPS, but receiving these benefits. The residual is then distributed equally to all CPS individuals.

[19] In table 2.1, these items are included as part of compensation (line 2) and subsequently deducted in line 25.

[20] Prior to 2005, TAXSIM-35 imputations are used for the child care tax credit.

[21] See section 7 for a more detailed explanation of this strategy.